3D Vectors: The Scalar Product

Questions 1–29

Question 1. Find the length of the following vectors:

a) $\mathbf{a} = \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}$

b) $\mathbf{b} = 12\mathbf{j} - 16\mathbf{k}$

Question 2. Given that the scalar product is distributive, find the value of:

a) $\mathbf{i} \cdot \mathbf{j}$

b) $\mathbf{j} \cdot \mathbf{k}$

c) $\mathbf{i} \cdot \mathbf{i}$

d) $5\mathbf{j} \cdot 3\mathbf{j}$

e) $2\mathbf{j} \cdot 6\mathbf{k}$

f) $\mathbf{i} \cdot (\mathbf{i} + \mathbf{j})$

g) $\mathbf{k} \cdot (2\mathbf{i} - \mathbf{j})$

h) $\mathbf{i} \cdot (3\mathbf{i} - 2\mathbf{j} + \mathbf{k})$

i) $(\mathbf{i} + \mathbf{j}) \cdot (\mathbf{i} + \mathbf{k})$

j) $(2\mathbf{i} - 3\mathbf{k}) \cdot (2\mathbf{j} - 3\mathbf{k})$

k) $(\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$

Question 3. Find the scalar product of:

a) $\mathbf{a} = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 10 \\ 20 \\ -5 \end{pmatrix}$

b) $\mathbf{a} = 6\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{b} = -3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$

c) $\mathbf{a} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$

d) $\mathbf{a} = 8\mathbf{i} - 15\mathbf{k}$ and $\mathbf{b} = 16\mathbf{i} - 18\mathbf{j}$

Question 4. Find the value of $(2\mathbf{i} + p\mathbf{j} - 3p\mathbf{k}) \cdot (3p\mathbf{i} - 2\mathbf{j} - \mathbf{k})$ in terms of p.

Question 5. It is given that $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}$, and that θ is the acute angle between $\mathbf{a}$ and $\mathbf{b}$. Find the exact value of $\cos\theta$.

Question 6. Use the scalar product to find the acute angle between the following pairs of vectors. Give your answer in degrees to 1 decimal place.

a) $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$

b) $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

c) $\mathbf{a} = \begin{pmatrix} \sqrt{2} \\ 3 \\ 2\sqrt{2} \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 5\sqrt{2} \\ 5 \end{pmatrix}$

d) $\mathbf{a} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$

Question 7. The acute angle between vectors $\mathbf{a}$ and $\mathbf{b}$ is θ. It is given that $\mathbf{a} \cdot \mathbf{b} = 8$, $|\mathbf{a}| = 2$ and $|\mathbf{b}| = \frac{11}{2}$. Find the exact value of $\sin\theta$.

Question 8. The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively relative to an origin O, where $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - \mathbf{j} - \mathbf{k}$. Find the angle OAB.

Question 9. The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively relative to an origin O, where $\mathbf{a} = -4\mathbf{i} + \mathbf{j} - 8\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$. The angle AOB is θ.

a) Find $\cos\theta$.

b) Find the area of the triangle AOB.

Question 10. It is given that $A(-3, -1, 2)$, $B(8, -4, 4)$ and $C(2, -9, -10)$. Find the angle ABC to 1 decimal place.

Question 11. In each case, determine if the following pairs of vectors are perpendicular.

a) $\mathbf{a} = \begin{pmatrix} 6 \\ -1 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ 4 \\ -5 \end{pmatrix}$

b) $\mathbf{a} = -2\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 6\mathbf{j} - 6\mathbf{k}$

c) $\mathbf{a} = \begin{pmatrix} 9 \\ -6 \\ 15 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -5 \\ -10 \\ -1 \end{pmatrix}$

Question 12. It is given that $A(8, -2, 1)$, $B(10, -6, 5)$, $C(10, -4, 2)$ and $D(12, 4, 5)$ are the vertices of a triangular-based pyramid. The angle BAC is θ.

a) Find the exact value of $\sin\theta$.

b) Show that $\overrightarrow{AD}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

c) Given that the volume of a pyramid is $\frac{1}{3}\times$ base area $\times$ perpendicular height, find the volume of the pyramid $ABCD$.

Question 13. It is given that $A(1, 9, -6)$, $B(2, 5, 2)$ and $C(\frac{11}{3}, \frac{28}{3}, \frac{7}{3})$ are the vertices of a triangle. D is the point on AB such that $AD : DB = 2 : 1$.

a) Show that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{DC}$.

b) Find the exact area of the triangle ABC.

Question 14. Determine the value(s) of p that would make the following pairs of vectors perpendicular:

a) $\mathbf{a} = 3p\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - 4p\mathbf{j} + 6\mathbf{k}$

b) $\mathbf{a} = \begin{pmatrix} 16 \\ -4p \\ 7 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 7 \\ 40p \end{pmatrix}$

c) $\mathbf{a} = 5p\mathbf{i} - p\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = p\mathbf{i} - 5\mathbf{j} + 8p\mathbf{k}$

d) $\mathbf{a} = \begin{pmatrix} 4p \\ -6p \\ 12 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3p \\ 2 \\ 6 \end{pmatrix}$

Question 15. The points A, B and C have position vectors $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$ and $\mathbf{c} = 6\mathbf{i} + (p-5)\mathbf{j} - \mathbf{k}$ respectively, where p is a constant.

a) Show that angle ACB can never be $90°$.

b) Find the value of p such that:

(i) angle $ABC = 90°$

(ii) A, B and C are collinear.

Question 16. It is given that $\mathbf{a} = \begin{pmatrix} 3k \\ -2 \\ 5 \end{pmatrix}$, where k is a constant, is perpendicular to $\mathbf{b} = \begin{pmatrix} 4 \\ -3 \\ -2 \end{pmatrix}$. Find the value of k.

Question 17. It is given that $\mathbf{a} = \begin{pmatrix} -6 \\ 2(k+1)^2 \\ 29 \end{pmatrix}$, where k is a constant, is perpendicular to $\mathbf{b} = \begin{pmatrix} 8 \\ -5 \\ 2 \end{pmatrix}$. Find the possible values of k.

Question 18. It is given that $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$. The vector $\mathbf{c} = i\mathbf{i} + m\mathbf{j} + n\mathbf{k}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, where m and n are constants. Find the values of m and n.

Question 19. It is given that $\mathbf{a} = \begin{pmatrix} -13 \\ -4 \\ -9 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ -7 \\ 10 \end{pmatrix}$. The vector $\mathbf{c} = \begin{pmatrix} m \\ n \\ 5 \end{pmatrix}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, where m and n are constants. Find the values of m and n.

Question 20. Find a non-zero vector that is perpendicular to both $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 4\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$.

Question 21. Find a unit vector that is perpendicular to both $\mathbf{a} = -42\mathbf{i} - \mathbf{j} + 6\mathbf{k}$ and $\mathbf{b} = 10\mathbf{i} + 5\mathbf{j} - 5\mathbf{k}$.

Question 22. Find a non-zero vector, in terms of p, that is perpendicular to both $\mathbf{a} = \begin{pmatrix} p \\ 3p \\ 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 4 \\ 3 \end{pmatrix}$.

Question 23. Find, in terms of the constant a, a vector that is perpendicular to both $\mathbf{a} = \begin{pmatrix} 2q \\ -5 \\ 3q \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$.

Question 24. Show that the vectors $\mathbf{a} = \begin{pmatrix} x+1 \\ x-2 \\ 2x+3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 4x \\ 7 \end{pmatrix}$ can never be perpendicular.

Question 25. Show that the vectors $\mathbf{a} = \begin{pmatrix} 3-x \\ 4-3x \\ 5-x \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} x-1 \\ x \\ -5 \end{pmatrix}$ can never be perpendicular.

Question 26. It is given that $\mathbf{a} = \begin{pmatrix} 2 \\ q \\ -3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} q+1 \\ q \\ -1 \end{pmatrix}$, where q is a constant.

a) Find a vector $\mathbf{c}$ that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

b) Given that $\mathbf{d} = \begin{pmatrix} -5q \\ 5 \\ 1 \end{pmatrix}$, show that $\mathbf{c}$ is never perpendicular to $\mathbf{d}$.

Question 27. Find the shortest possible vector of the form $\begin{pmatrix} m \\ n \\ 1 \end{pmatrix}$ that is perpendicular to both $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ x \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -4 \\ 3 \\ x+5 \end{pmatrix}$.

Question 28. Find the shortest possible vector of the form $\begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ that is perpendicular to both $\mathbf{a} = \begin{pmatrix} 1 \\ x \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 1-x \\ 8 \end{pmatrix}$.

Question 29. It is given that $\mathbf{p}$ and $\mathbf{q}$ are nonzero 3D vectors. It is also given that $\mathbf{p}+\mathbf{q}$ is perpendicular to $\mathbf{p}-\mathbf{q}$. Prove that $|\mathbf{p}| = |\mathbf{q}|$.